Problem 9.9

Eric deposits X into a savings account at time 0, which pays interest at nominal annual rate i, compounded semiannually. Mike deposits $2X$ into a different savings account at time 0, which pays simple interest at annual rate i. Eric and Mike earn the same amount of interest during the last six months of the eighth year. Calculate i.

Solution

Since Eric's account is compounded semiannually, the interest rate per half-year is

$$j = \frac{i}{2}.$$

The last six months of the eighth year correspond to the period from time 7.5 to time 8. Before this period, Eric's account has accumulated for 7.5 years, or 15 half-year periods. Therefore, Eric's interest during this last half-year is

$$I_E = X\left(1+\frac{i}{2}\right)^{15}\left(\frac{i}{2}\right).$$

Mike earns simple interest at annual rate i on the initial deposit $2X$. Thus, his interest during any half-year period is

$$I_M = 2X\left(\frac{1}{2}\right)i = Xi.$$

The problem states that both amounts of interest are equal, so

$$X\left(1+\frac{i}{2}\right)^{15}\left(\frac{i}{2}\right) = Xi.$$

Canceling Xi from both sides gives

$$\frac{1}{2}\left(1+\frac{i}{2}\right)^{15} = 1.$$

Hence,

$$\left(1+\frac{i}{2}\right)^{15} = 2.$$

Taking the fifteenth root of both sides,

$$1+\frac{i}{2} = 2^{1/15}.$$

Therefore,

$$i = 2\left(2^{1/15}-1\right).$$

Numerically,

$$i \approx 0.0945882.$$

Thus, the nominal annual interest rate is

$$\boxed{i \approx 9.46\%}.$$